Capstone Project Proposal

Early Churn Prediction for Subscription-Based Digital Services Using Time-Aware Machine Learning Pipelines

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1. Problem Statement

Customer churn is a critical business challenge for subscription-based digital services such as music streaming, video platforms, SaaS products, and cloud services. Retaining existing customers is often significantly more cost-effective than acquiring new ones, yet many organizations struggle to identify early signals of disengagement before a customer cancels their subscription.

The objective of this project is to build a **time-aware machine learning system** that predicts whether a user is likely to churn based on their **early behavioral and transactional patterns**. The model will generate churn risk scores that could be used by product, marketing, or customer success teams to proactively intervene with retention strategies.

This project focuses not only on predictive performance, but also on **engineering practices required to deploy and maintain churn models in real-world production environments**, including reproducibility, scalability, and monitoring considerations.

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2. Dataset Description and Data Collection

The primary dataset for this project is the **KKBox Churn Prediction Dataset**, originally released as part of a Kaggle machine learning challenge. KKBox is a music streaming service, and the dataset reflects real-world subscription and user activity data.

Key characteristics of the dataset include:

- **User activity logs** capturing listening behavior over time
- **Transaction data** including subscription renewals and cancellations
- **Membership metadata** such as plan types and tenure

- **Large scale** (multiple gigabytes, millions of records across multiple tables)
- **Temporal structure**, enabling time-based feature engineering and evaluation

The dataset is well-suited for this project because it is:

- Clean and well-documented
- Structured across multiple relational tables
- Large enough to simulate production-scale challenges
- Representative of real-world churn prediction problems

Due to dataset size and licensing constraints, the full dataset will be stored locally and accessed via documented download instructions. Small, reproducible sample extracts will be included in the project repository for inspection and experimentation.

To supplement exploratory analysis and feature ideation, two smaller public datasets (Telco Customer Churn and Online Retail II) may be used for comparison and validation of churn-related feature patterns.

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3. Proposed Approach and Methodology

The project will follow an **end-to-end machine learning engineering workflow**, including:

1. **Data ingestion and preprocessing**
    - Joining user activity, transaction, and membership data
    - Handling missing values and categorical encoding
    - Creating time-based snapshots using a fixed prediction cutoff to avoid data leakage
2. **Feature engineering**
    - Behavioral features (recency, frequency, inactivity patterns)
    - Transactional features (renewal behavior, payment gaps)

    - Aggregated temporal features using rolling windows

3. **Model development**
    - Baseline model (logistic regression)
    - Tree-based models such as LightGBM and XGBoost
    - Use of sklearn pipelines for reproducibility
4. **Evaluation strategy**
    - Time-based train/validation splits
    - Metrics relevant to churn use cases (ROC-AUC, precision-recall, lift at top-k)
    - Analysis of feature importance and model stability
5. **Production-readiness considerations**
    - Configuration-driven training
    - Model artifact versioning
    - Basic drift and performance monitoring simulations across time windows

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4. Ethical Considerations

Churn prediction systems can influence how users are treated by a business, which raises several ethical considerations:

- **Bias and fairness:** Certain demographic or behavioral patterns could unintentionally lead to biased predictions. Care will be taken to evaluate features for potential proxy bias and to avoid using sensitive attributes directly.
- **Transparency:** The model will emphasize interpretability through feature importance analysis to support responsible use.
- **User impact:** Churn scores should be used to improve user experience and engagement, not to penalize or restrict users.

The project will explicitly document these considerations and discuss how responsible model usage can mitigate negative consequences.

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5. Computational Resources and Feasibility

The project is designed to be computationally feasible on a personal development environment:

- **Hardware:** Local machine with standard CPU resources (no GPU required)
- **Memory:** Moderate memory usage through efficient sampling and batch processing
- **Tools:** Python, pandas, scikit-learn, LightGBM/XGBoost
- **Data handling:** Large datasets will be processed incrementally or via filtered subsets where appropriate

This ensures the project remains within the scope of the course while still addressing real-world scale challenges.

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6. Project Significance and Domain Relevance

This project addresses a **high-impact, real-world problem** that is directly relevant to modern digital products and subscription-based businesses. It leverages the author's professional background in CRM systems, customer lifecycle analytics, and enterprise software architecture to frame churn prediction not just as a modeling task, but as a system that must operate reliably in production.

By emphasizing time-aware modeling, reproducibility, and monitoring considerations, this project aligns strongly with **Machine Learning Engineering and MLOps roles**, while also demonstrating product-level thinking relevant to AI-driven decision-making systems.

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7. Expected Outcomes

By the end of this project, the expected outcomes include:

- A reproducible churn prediction pipeline

- A trained and evaluated machine learning model
- Clear documentation of data handling, modeling decisions, and ethical considerations
- A GitHub repository suitable for technical review and discussion in interviews